Working Paper # 30

Revision of Program - BE Software

Sponsored by SEECS

Background

1. The aim of this working paper is to make the BESE curriculum in line with PEC Engineering Education Guidelines issued for the adoption of HEC UEP-2024 (HEC letter No. HEC/Acad/UGEP/2024/6537 dated 19-09-2024 & HEC letter No. 10-01/2022/Coord (Acad)/HEC/235 dated March 28, 2025).

Rationale

2. The Pakistan Engineering Council (PEC) has issued guidelines (letter no. PEC/AD/CD/2025/03 dated April 10, 2025) for incorporating the Higher Education Commission's Undergraduate Education Policy 2024 (HEC UEP-2024) into undergraduate programs. The HEC and the PEC Engineering Accreditation Board have mandated the inclusion of compulsory courses—Pakistan Studies, Fehm-e-Quran-I, and Fehm-e-Quran-II—within the General Education/Non-Engineering domain of all engineering programs, effective from Fall 2025. To meet these requirements, it is proposed to revise the Software Engineering curriculum in accordance with the PEC Framework.

Educational Objectives

3. SE curriculum compliance with PEC guidelines incorporating HEC UEP- 2024.

Proposed Changes

4. As per PEC's proposed guideline:
    a. Mandatory course of "Pakistan Studies" (2+0 Cr Hrs) will be offered in the fourth semester.
    b. Two courses "Understanding of Holy Quran-I/ Fehm-e-Quran-I" and "Understanding of Holy Quran-II/ Fehm-e-Quran-II" (each of 1 Cr Hr) will be offered in the first and second semester, respectively.
    c. "Applications of ICT" (2+1 Cr Hrs) will be offered in the first semester.
    d. To comply with the minimum credit hour requirement of 15 per semester set by HEC and PEC, "Occupational Health and Safety" and "Sociology Elective-II" have been moved to the eighth semester. This adjustment

ensures a balanced distribution of credit hours without disrupting the sequence of core Software Engineering courses. To remain within the maximum credit hour limit of 21 per semester, few courses have been relocated.

e. As part of the ongoing curriculum review, the Departmental Board of Studies has assessed the alignment of current course offerings with the Pakistan Engineering Council (PEC) guidelines and the evolving academic and industry requirements of the Software Engineering discipline. Based on this evaluation, it is proposed that the courses “Engineering Drawing” and “Workshop Practice” be removed from the undergraduate Software Engineering curriculum. These courses were originally included as part of a common first-year structure across all engineering programs; however, they are no longer a requirement under the current PEC curriculum for Software Engineering. Therefore, their removal ensures greater relevance and compliance with updated accreditation standards. In addition to enhancing the curriculum's relevance, the removal of these non-essential courses will also support compliance with the maximum credit hour limit of 140, as prescribed by HEC and PEC for undergraduate engineering programs. This adjustment will enable the inclusion of more domain-specific and industry-relevant courses while maintaining regulatory compliance. The approved curriculum currently offered is shown against the proposed SE Curriculum in Annex A.

f. The mapping of elective courses in semester-wise breakdown of BESE curriculum shall remain dynamic. Electives can be offered with 3-1, 3-0 or 2-1 credit hours. List of elective courses is given in Annex B.

g. To ensure alignment with PEC guidelines, the number of core engineering elective courses has been increased from five to six. Corresponding adjustments have been made to the credit hours allocated for elective courses to accommodate this revision.

Approval by DBS/FBS

5. Approved in FBS held on 8th July 2025. Meeting minutes are attached in Annex C.

Comments of Academics Directorate

6. The proposal was deliberated and endorsed by UCRC held on 28 Aug 2025.

Recommendation of Academics Directorate

7. Revision of program- unified BE Software is recommended for approval w.e.f Fall 2025 at SEECS & MCS.

8. Academic council is requested for the decision.

Prepared by: Ms. Mehwish Kiran

Date:30th June 2025

Checked by**: Professor Dr. Rafia Mumtaz (Assoc Dean FoC),**

Dr. Momina Moetesum (HoD SE)

Approved by: **Dr. Ajmal Khan (Principal SEECS)**

Date: 23rd July 2025

Annex-A

Approved Scheme in the 69th ACM				Proposed Curriculum			
Semester I				**Semester I**			
Course Code	**Subjects**	**Credit Hrs**		**Course Code**	**Subjects**	**Credit Hrs**	
HU-114	Functional English***	3-0		HU-XXX	Fehm-e-Quran-I*	0-1	
CS-110	Fundamentals of Computer Programming	3-1		CS-110	Fundamentals of Computer Programming	3-1	
HU-127	Ideology and Constitution of Pakistan	2-0		HU-127	Ideology and Constitution of Pakistan	2-0	
MATH-101	Calculus and Analytical Geometry	3-0		MATH-101	Calculus and Analytical Geometry	3-0	
PHY-102	Applied Physics	2-1		PHY-102	Applied Physics	2-1	
MATH-161	Discrete Mathematics***	3-0		HU-101/ HU-XXX	Islamic Studies/ Ethics***	2-0	
				CS-117	Applications of ICT*	2-1	
	Total CHs	**16-2 (18)**			**Total CHs**	**14-4 (18)**	
Semester II				**Semester II**			
CS-212	Object Oriented Programming (OOP)	3-1		CS-212	Object Oriented Programming (OOP)	3-1	
OHS-101	Occupational Health and Safety***	1-0		HU-XXX	Fehm-e-Quran-II*	0-1	
MATH-121	Linear Algebra and ODEs	3-0		MATH-121	Linear Algebra and ODEs	3-0	
ME-109	Engineering Drawing**	0-2		MATH-161	Discrete Mathematics***	3-0	
ENGL-104	Expository Writing***	3-0		HU-114	Functional English***	3-0	
EE-122	Computer Architectureand Logic Design	3-1		EE-122	Computer Architecture and Logic Design	3-1	
ME-105	Workshop Practice**	0-1					
	Total CHs	**13-5 (18)**			**Total CHs**	**15-3 (18)**	

Semester III				Semester III			
CS-220	Database Systems	3-1		CS-220	Database Systems	3-1	
SE-200	Software Engineering	3-0		SE-200	Software Engineering	3-0	
CS-250	Data Structures & Algorithms	3-1		CS-250	Data Structures & Algorithms	3-1	
MATH-352	Numerical Methods***	2-1		MATH-232	Complex Variables andTransform***	3-0	
EE-353	Computer Networks	3-1		EE-353	Computer Networks	3-1	
	Total CHs	**14-4 (18)**			**Total CHs**	**15-3 (18)**	
Semester IV				**Semester IV**			
CS-330	Operating Systems	3-1		CS-330	Operating Systems	3-1	
CS-251	Design & Analysis of Algorithm	3-0		CS-251	Design & Analysis of Algorithm	3-0	
MATH-232	Complex Variables andTransform	3-0		HU-107	Pakistan Studies*	2-0	
SE-211	Software Design and Architecture	2-1		SE-211	Software Design and Architecture	2-1	
MATH-361	Probability and Statistics	3-0		MATH-361	Probability and Statistics	3-0	
HU-101/ HU-XXX	Islamic Studies/ Ethics***	2-0		ENGL-104	Expository Writing***	3-0	
	Total CHs	**16-2 (18)**			**Total CHs**	**16-2 (18)**	
Semester V				**Semester V**			
	Management ScienceElective – I	3-0			Management Science Elective – I	3-0	
	Engineering Elective –I	3-1			Engineering Elective – I	3-X	
SE-315	Cloud Computing	2-1		SE-315	Cloud Computing	2-1	
	MDEE – I	2-1			MDEE – I	2-1	
	Sociology Elective – I	2-0			Sociology Elective – I	2-0	
SE-314	Software Construction	2-1		SE-314	Software Construction	2-1	
	Total CHs	**14-4 (18)**			**Total CHs**	**14-(3+X) (17+X)**	
Semester VI				**Semester VI**			
SE-320	Formal Methods	3-0		SE-320	Formal Methods	3-0	
SE-321	Software Quality Engineering	3-0		SE-321	Software Quality Engineering	3-0	

	Management Science Elective – II	2-0			Management Science Elective – II	2-0	
	Engineering Elective-II	3-X			Engineering Elective-II	3-X	
SE-430	Software Project Management	3-0		SE-430	Software Project Management	3-0	
CS-261	Human Computer Interaction	2-1		CS-261	Human Computer Interaction	2-1	
	Total CHs	**16-(1+X) (17+X)**			**Total CHs**	**16-(1+X) (17+X)**	
Semester VII				**Semester VII**			
SE-498	Final Year Design Project – I	0-2		SE-498	Final Year Design Project – I	0-2	
	Engineering Elective –III	3-X			Engineering Elective – III	2-1/3-0	
	MDEE – II	3-0			MDEE – II	3-0	
	Sociology Elective – II	2-0		MATH-352	Numerical Methods***	2-1	
CS-484	Information Security	3-0		CS-484	Information Security	3-0	
					Engineering Elective – IV	2-1/3-0	
	Total CHs	**11-(2+X) (13+X)**			**Total CHs**	**12-5 /14-3 (17)**	
Semester VIII				**Semester VIII**			
SE-499	Final Year Design Project – II	0-4		SE-499	Final Year Design Project – II	0-4	
	Engineering Elective –IV	3-1			Engineering Elective – V	2-1/3-0	
CCE-401	Civic and Community Engagement	1-1		CCE-401	Civic and Community Engagement	1-1	
	Engineering Elective –V	3-1			Engineering Elective –VI	2-1/3-0	
				OHS-101	Occupational Health and Safety***	1-0	
					Sociology Elective – II***	2-0	
	Total CHs	**7-7 (14)**			**Total CHs**	**8-7 /10-5 (15)**	
	Grand Total	**107+27+X (134+X)**			**Grand Total**	**138+X**	

*New Courses to be introduced as per PEC Guidelines

**Courses removed from SE curriculum

***Inter-semester shuffling of existing courses

Annex-B

Software Eng- Core Depth Elective Courses **Note: No limit on number of courses, minimum 18 Credit Hours**			
S.No	**Course Code**	**Course Name**	**Credit Hours**
1	CS-213	Advanced Programming	4(3+1)
2	CS-321	Advanced Database Systems	3(3+0)
3	CS-322	RDBMS Using Oracle	3(2+1)
4	CS-332	Distributed Computing	4(3+1)
5	CS-340	Web Technologies I	3(2+1)
6	CS-342	Mobile Computing	3(3+0)
7	CS-344	Web Engineering	4(3+1)
8	CS-352	Theory of Automata and Formal Languages	3(3+0)
9	CS-361	Computer Graphics	4(3+1)
10	CS-362	Multimedia Systems and Design	3(2+1)
11	CS-363	Visualization	3(2+1)
12	CS-364	Game Programming	3(2+1)
13	CS-370	Artificial Intelligence	4(3+1)
14	CS-380	Introduction to Computer Security	3(3+0)
15	CS-382	Fundamentals of Cryptography	3(3+0)
16	CS-414	Advanced Java with emphasis on Internet Applications	4(3+1)
17	CS4-23	Data warehousing and Data mining	4(3+1)
18	CS-424	Information Retrieval	3(3+0)
19	CS-425	Management Information systems	3(3+0)
20	CS-433	Applied parallel Computing	3(2+1)
21	CS-441	Web Technologies-II	4(3+1)
22	CS-443	E-Commerce and Solutions	3(3+0)
23	CS-453	Programming Languages	3(3+0)
24	CS-471	Machine Learning	4(3+1)
25	CS-472	Natural Language Processing	3(3+0)
26	CS-474	Computer Vision	3(2+1)
27	CS-476	Speech and Image Processing	4(3+1)
28	CS-481	Computer Forensics	4(3+1)

29	CS-482	System Incident Handling	3(3+0)
30	CS-490	Advance Topics in Computing	3(3+0)
31	EC-303	Mobile Application Development for SME's	3(2+1)
32	EE-433	Digital Image Processing	4(3+1)
33	SE-301	Object Oriented Software Engineering	3(3+0)
34	SE-313	Design Patterns	3(2+1)
35	SE-422	Software Testing	3(3+0)
36	SE-423	Software Metrics	3(3+0)
37	SE-431	Software Engineering Economics	3(3+0)
38	SE-440	Business Process Automation	3(3+0)
39	SE-490	Advanced Topics in Software Engineering	3(3+0)
40	BIO-215	Bioinformatics	3(3+0)
41	BIO-317	Computational Biology	3(3+0)
42	CS-260	Human Computer Interaction	3(3+0)
43	CS-271	Computational Logic	3(3+0)
44	CS-331	System Programming	2(2+0)
45	CS-405	Deep Learning	4(3+1)
46	CS-381	Network Security	3(3+0)
47	CS-473	Theory Of Intelligent Systems	3(3+0)
48	EE-215	Electronic Devices and Circuits	4(3+1)
49	EE-232	Signals And Systems	3(3+0)
50	EE-414	Digital Electronics	4(3+1)
51	CS-416	Large Language Models	4(3+1)
52	SE-XXX	Financial Technologies	3 (2+1)
53	EE-232	Signals And Systems	3(3+0)
54	SE-410	System Modeling and Simulation	3(3+0)
55	CEM-300	Procurement Management	3(3+0)
56	EE-423	Embedded System Design	4(3+1)
57	EE-102	Basic Electrical Engineering	4(3+1)
58	EE-477	Analog and Digital Communication	4(3+1)
59	EE-330	Digital Signal Processing	4(3+1)
60	CS-334	Open-Source Systems	4(3+1)
61	CS-XXX	Agent Based Software Engineering	3(2+1)
62	CS-404	Big Data Analytics	3(2+1)

63	CS-XXX	Computer Graphics	3(2+1)
64	CS-XXX	Visual Programming	3(2+1)
65	CS-XXX	Real Time Systems	3(2+1)
66	CS-XXX	Data Encryption and Security	3(2+1)
67	CS-XXX	Deep Learning	3(2+1)
68	CS-XXX	Network Security	3(2+1)
69	CS-XXX	Virtual and Augmented Reality	3(2+1)
70	CS-XXX	Reinforcement Learning	3(2+1)
71	CS-415	Mobile Application Development	4(3+1)
72	CS-XXX	Large Language Models	3(2+1)
73	CS-XXX	Computer Forensics	3(2+1)
74	CS-245	Machine Learning	3(2+1)
75	SE-XXX	Micro-services design and development	3 (2+1)
76	IS-241	Network Security Engineering	4(3+1)
77	IS-379	Digital Information Forensic (2+1)	3(2+1)
78	IS-382	Security Engineering Management (3+0)	3(3+0)
79	CS-XXX	Natural Language processing	4 (3+1)
80	CS-XXX	Reinforcement Learning	4(3+1)
81	CS- XXX	Artificial Intelligence	3 (2+1)
82	CS-475	Computer Vision	3 (3+0)
83	CS-416	Natural language processing	3 (2+1)
84	EC-312	Digital Image Processing	3 (2+1)
85	CS-3XX	Web Engineering	3 (2+1)
87	CS-3XX	Data Mining	3 (2+1)
88	CS-xxx	Advanced Programming	3 (2+1)
89	CS-347	Parallel and Distributed Computing	3 (2+1)
90	CS-470	Machine Learning	3 (3+0)

Multi-Disciplinary Engineering Electives Note: No limit on number of courses, minimum 6 Credit Hours			
S.No	**Course Code**	**Course Name**	**Credit Hours**
1	CS-336	Embedded Systems	3(2+1)
2	CS-335	Internet of Things	3(3+0)
3	CS-382	Fundamentals of Cryptography	3(3+0)
4	EE-201	Engineering Mechanics	3(3+0)
5	MATH-133	Engineering Mathematics	3(3+0)

6	MATH-221	Number Theory	3(3+0)
7	PHY-401	Advanced Physics	3(2+1)
8	EE-212	Basic Electronics	3(2+1)
9	EE-322	Wireless Networks	3(3+0)
10	EE-350	Data Communication	3(3+0)
11	EE-430	Telecommunication Systems	3(3+0)
12	EE-430	Introduction To Robotics (3+0)	(3+0)
13	IS-353	Cryptographic Design Engineering (3+1	(3+1)
14	ME-446	Computer-Aided Engineering (2+1)	(2+1)
15	PHY-434	Quantum Computation	(3+0)

Non-Engineering Electives
Note: No limit on number of courses, minimum 9 Credit Hours

Note: No limit on number of courses, **minimum 9 Credit Hours** (The list below is by no means exhaustive. Any NUST ACM Approved University Elective can be opted with the approval of Department)

Management Electives (5 CHs)

Ser	Course Code	Course Name	Credit Hours
1	FIN-100	Principles of Accounting	3(3-0)
2	FIN-204	Financial Management for IT Professional	2(2-0)
3	MGT-164	Introduction to Management	2(2-0)
4	MGT-175	Intellectual Property Rights	3(3-0)
5	MGT-452	New Business Ventures	2(2-0)
6	MKT-102	Principles of Business and Marketing	2(2-0)
7	OTM-455	Engineering Project Management	2(2-0)
8	MGT-271	Entrepreneurship	2(2-0)

Social Science Elective (4 CHs)

S.No	Course Code	Course Name	Credit Hours
1	CS-309	Computing and Society	3(3-0)
2	ECO-130	Engineering Economics	2(2-0)
3	HRM-240	Organizational Behavior	2(2-0)
4	HRM-241	Organizational Behavior	3(3-0)
5	HRM-441	Human Resource Management	2(2-0)
6	HRM-442	Human Resource Management	3(3-0)
7	HU-102	Psychology	3(3-0)
8	HU-103	Principles of Sociology	3(3-0)
9	HU-223	Professional Ethics	3(3-0)